

Maddie Burroughs

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EDUCATION

Carnegie Mellon University

Bachelor of Science in Electrical and Computer Engineering

Additional Major in Design for Interactive Media

May 2025

Pittsburgh, PA

EXPERIENCE

Origami Risk

Aug 2025 – Present

Associate Full Stack Software Engineer

Chicago, IL

- Develop and maintain security-focused product features and fixes across the application, addressing vulnerabilities surfaced through audits, customer reports, QA findings, and internal testing.
- Investigate and remediate security issues including XSS vulnerabilities, input validation gaps, and scaling-related architectural design flaws across a rapidly growing enterprise software platform.
- Design, implement, and maintain robust custom validation tools to enable safe execution of customer-authored JavaScript and SQL code.
- Designed and built a microservice for codeless configuration of payment providers, generalizing API steps and data structures into a block-based workflow interface for non-technical users.

Carnegie Mellon University

Aug 2022 – May 2025

Content Lead, Events Lead – Fundamentals of Programming (15-112/110)

Pittsburgh, PA

- Led curriculum development for a 500 student introductory programming course, reviewing, approving, and writing recitation materials, assignments, practice assessments, and supplemental tutoring content.
- Redesigned the course's month-long term project, improving assignment structure, scaffolding, and instructional clarity for beginner programmers.
- Taught twice-weekly recitations, hosted office hours and small-group tutoring sessions, and supported grading and student feedback across core programming topics.
- Planned and hosted large-scale course events for 200+ students, including a 24-hour hackathon designed to build community and encourage collaborative application of course concepts.
- Collaborated with CS Academy staff to iterate on online course content and improve the digital learning platform used to deliver interactive programming instruction.

Shirley Ryan AbilityLab – Center for Bionic Medicine

May 2023 – Aug 2023

Mechatronics Engineering Intern

Chicago, IL

- Developed a Python-based prosthetic sensor signal visualization GUI to support real-time data analysis during physical therapy sessions.
- Built and tested electrical components for ankle and knee prosthetics, supporting ongoing development and validation of bionic mobility devices.
- Updated and maintained an upper-limb prosthetic development kit, improving usability and reliability for research and prototyping workflows.

Scotch'n'Soda Theatre Company

Aug 2021 – May 2025

Production Manager, Technical Director, Designer

Pittsburgh, PA

- Scheduled and facilitated seven rounds of deliverable meetings between designers, implementers, and technical leads across scenic, lighting, sound, costuming, hair and makeup, and publicity teams.
- Managed production budgets of up to \$25,000, tracking resource needs across technical and creative teams.
- Coordinated communication between technical and creative teams, production leadership, and university stakeholders throughout production cycles lasting up to five months.
- Facilitated full-company production events for up to 125 people, including design previews, load-ins, strikes, and multiple safety inspections.

SKILLS

Languages	Python, C#, JavaScript, TypeScript, SQL, HTML/CSS, C++, C, Java, SystemVerilog, MATLAB
Frameworks & Platforms	React, .NET, Django, AWS, Docker, Auth0, LaunchDarkly, Temporal, Arduino, OpenCV
Developer Skills & Tools	Git/GitHub, LaTeX, SQL Server, Database Management, Agile, API Design, Security Testing, Vulnerability Remediation
Design & CAD	Figma, Photoshop, Illustrator, AutoCAD, SolidWorks
Fabrication & Prototyping	Technical Drafting, Construction Plan Review, Circuit and PCB Design, Soldering, FDM/SLA 3D Printing, Laser Cutting, Woodworking, Sewing, Glassblowing, Ceramics